25901106 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBTS201 - RESEARCH METHODOLOGY AND BIOSTATISTICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. A population census that systematically describes demographic characteristics is an example of: [U] / [CO1]

   a). Exploratory research b). Action research

   c). Descriptive research d). Basic research

2. Grouping people as Male and Female is an example of: [U] / [CO2]

   a). Quantitative classification b). Chronological classification

   c). Qualitative classification d). Geographical classification

3. ANOVA test statistic is: [U] / [CO3]

   a). t b). Z

   c). F d). Chi square

4. The discussion section of a research report is used to: [U] / [CO4]

a). Repeat the abstract

b). List all sources alphabetically

c). Define research tools

d). Interpret and explain the findings

5. Which of the following is an example of a limitation in research? [U] / [CO2]

a). Restricting the study to one city

b). Conducting increased number of sampling

c). Limited sample size due to time and budget constraints

d). Using surveys to collect data

6. An alternative hypothesis (H_1) indicates that: [U] / [CO2]

a). No relationship exists

b). The research is invalid

c). A significant relationship or difference exists

d). Variables are constant

7. Which method of data collection covers the widest area: [U] / [CO2]

a). Mailed questionnaire method

b). Direct interview method

c). Telephone interview

d). none of the above

8. Which one is a random sampling? [U] / [CO3]

a). Simple Random sampling

b). Quota sampling

c). Convenience sampling

d). Purposive sampling

9. A project proposal is mainly prepared to: [U] / [CO5]

a). Request approval or funding for a project

b). Publish research findings

c). Record daily activities

d). Give public awareness on a subject

10. In which method does the investigator fill the answers? [U] / [CO2]

a). Questionnaire

b). Schedule

c). Observation

d). Experiment

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. Define bibliography and explain its purpose. [U] / [CO2]

12. Explain the role of citation indexes in academic research and publication. [U] / [CO5]

13. Define sample space. [U] / [CO3]

14. What is budget justification in a research proposal? [U] / [CO5]

15. Explain paired t-test. [U] / [CO3]

16. List out the steps to define a research problem. [K] / [CO2]

17. What are the main characteristics of research? [K] / [CO1]

18. Explain the difference between results and discussion sections in a journal paper. [U] / [CO5]

19. Briefly explain simple bar diagram. [U] / [CO2]

20. What is secondary data? Give two examples of secondary data. [U] / [CO2]

21. What is a conference? explain its main purpose. [U] / [CO5]

22. Briefly explain SPSS. [U] / [CO3]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Write an essay on a research paper and its significance in research communication. [U] / [CO5]

24. Write a short essay on interpretation with empahsise to its techniques and precautions to be taken. [U] / [CO2]

25. Distribution of MBBS students according to blood group is as follows. Represent the data using simple bar diagram. [A] / [CO3]

Blood group	Number of students
A	16
B	4
AB	10
O	20

26. Explain the need of a good research design and its features to execute a proper research. [U] / [CO2]

27. What are the principles of good editing? [U] / [CO2]

28. Explain steps in hypothesis testing. [U] / [CO3]

29. Distinguish between discrete and continuous frequency table. [U] / [CO3]

30. Write a short essay on various types of research reports. [U] / [CO4]

[5x6 = 30]